

# Amazon FSx for NetApp ONTAP – Complete Introduction and Architecture Guide

## Module Objectives

After completing this module, you will understand:

- What Amazon FSx for NetApp ONTAP is
- Key benefits of FSx for ONTAP
- FSx architecture and high availability design
- Storage efficiency features
- Capacity Pool Tiering
- Disaster Recovery options
- Multi-Protocol access (NFS + SMB)
- Data migration methods
- Basic administration using ONTAP CLI
- Common use cases and interview questions

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## 1. What is Amazon FSx for NetApp ONTAP?

Amazon FSx for NetApp ONTAP (FSxN) is a fully managed AWS storage service built on NetApp ONTAP technology.

Instead of purchasing, deploying, and managing physical storage infrastructure, AWS manages:

- Hardware
- Storage controllers
- ONTAP upgrades
- Patching
- Backups
- High Availability operations

This allows administrators to focus on data management rather than infrastructure management.

## Traditional ONTAP vs FSx for ONTAP

| Traditional NetApp | FSx for ONTAP |
|--------------------|---------------|
| Purchase hardware  | AWS managed   |
| Install ONTAP      | Pre-installed |
| Manage HA pairs    | AWS managed   |
| Upgrade ONTAP      | AWS managed   |
| Configure disks    | AWS managed   |

| Traditional NetApp | FSx for ONTAP |
|--------------------|---------------|
| Monitor hardware   | AWS managed   |

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## 2. Key Benefits of FSx for ONTAP

### Fully Managed Service

AWS manages:

- Storage hardware
- Disk replacement
- Controller failover
- ONTAP updates
- Backup infrastructure

### AWS Native Integration

Works seamlessly with:

- Amazon EC2
- Amazon EKS
- Amazon ECS
- AWS Direct Connect
- AWS VPN
- AWS Backup

### Rapid Deployment

File systems can be deployed within minutes instead of weeks.

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## 3. ONTAP Features Available in FSx

FSx includes many enterprise ONTAP capabilities.

### Storage Efficiency

#### Deduplication

Eliminates duplicate blocks.

Example:

If 100 virtual machines contain identical operating system files, ONTAP stores only one copy.

Benefits:

- Reduced storage consumption
  - Lower costs
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## **Compression**

Compresses data before storing.

Benefits:

- Reduced capacity usage
  - Better storage utilization
- 

## **Compaction**

Packs partially used blocks together.

Benefits:

- Improved efficiency for small files
- 

## **Thin Provisioning**

Allocates storage logically before physical consumption.

Benefits:

- Better capacity utilization
  - Faster provisioning
- 

# **4. Capacity Pool Tiering**

## **Concept**

FSx contains two storage tiers:

### **SSD Tier**

High performance

Stores:

- Active data

- Frequently accessed data

### Capacity Pool Tier

Low-cost elastic storage

Stores:

- Cold data
  - Infrequently accessed data
- 

## Tiering Flow

```
Hot Data
↓
SSD Performance Tier
↓
Automatic Tiering Policy
↓
Capacity Pool Tier
```

Benefits:

- Lower storage cost
  - Automatic data movement
  - Elastic scaling
- 

## 5. FSx for ONTAP High Availability Architecture

### Multi-AZ Deployment

Each file system contains:

```
Availability Zone A
-----
Primary Node

    ||
    || Synchronous Replication
    ||

Availability Zone B
-----
Standby Node
```

Configuration:

- Active-Passive
  - Synchronous replication
  - Automatic failover
- 

## Failure Scenario

### Normal Operation

```
graph TD; Clients --> PrimaryNode[Primary Node]
```

Clients  
↓  
Primary Node

### Controller Failure

```
graph TD; Clients --> StandbyNode[Standby Node]
```

Clients  
↓  
Standby Node

Failover is automatic and transparent.

After recovery:

```
graph TD; StandbyNode --> Failback[Automatic Failback] --> PrimaryNode[Primary Node]
```

Standby Node  
↓  
Automatic Failback  
↓  
Primary Node

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## 6. Supported Protocols

FSx supports:

### NAS Protocols

#### NFS

Used by:

- Linux
- Unix

Versions:

- NFSv3
  - NFSv4
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## **SMB**

Used by:

- Windows
- Mac OS

Supports Active Directory integration.

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## **SAN Protocols**

### **iSCSI**

Provides block storage access.

Used for:

- Databases
  - Application servers
  - Virtualization workloads
- 

## **7. Multi-Protocol Access**

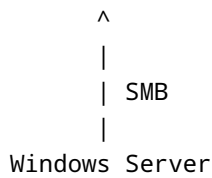
One of ONTAP's most powerful features.

Same data can be accessed simultaneously using:

- Linux via NFS
- Windows via SMB

Example:

```
Linux Server
  |
  | NFS
  |
  V
FSx Volume
```



Both clients see the same files.

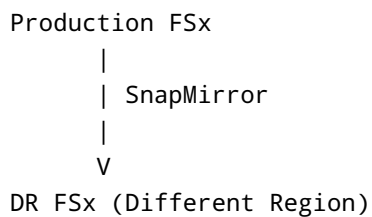
## 8. Disaster Recovery Options

### SnapMirror

Replication technology for:

- On-Prem NetApp → AWS
- AWS → AWS
- Cross Region DR

Example:



Benefits:

- Low RPO
- Fast recovery

### SnapVault

Used for:

- Backup
- Long-term retention

### FSx Backups

AWS managed backup solution.

Supports:

- Automatic backups
  - Manual backups
- 

## 9. Global Access Technologies

### FlexCache

Creates cached copies closer to users.

Example:

```
US Region
  |
  | Origin Volume
  |
  V

India Region
  |
  | FlexCache
  |
  Users
```

Benefits:

- Reduced latency
  - Faster file access
- 

### Global File Cache

Used for:

- Branch offices
- Remote users

Benefits:

- Local file access experience
  - Centralized data storage
-



## 10. Common FSx Use Cases

### Cloud Disaster Recovery

```
On-Prem NetApp
|
SnapMirror
|
V
FSx for ONTAP
```

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### Backup & Archive

```
Production Storage
|
SnapVault
|
V
FSx Backup Storage
```

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### Cloud Bursting

Use AWS compute resources while data remains on-premises.

```
On-Prem ONTAP
|
FlexCache
|
V
AWS Compute
```

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### Test & Development

Quickly clone production data.

Uses:

- FlexClone
  - Snapshots
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# 11. Creating an FSx File System

## Quick Create

Recommended for beginners.

Automatically creates:

- File System
- SVM
- Volume

Requires:

- File System Name
  - Capacity
  - VPC Selection
- 

## Standard Create

Allows customization of:

- Performance
  - Networking
  - Security
  - Backups
  - Maintenance windows
- 

# 12. Administrative Access

FSx provides:

## File System Management Endpoint

Used for:

- SSH access
  - ONTAP CLI administration
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## 13. Useful ONTAP Commands

### Show SVMs

```
vserver show
```

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### Show Volumes

```
volume show
```

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### Show Network Interfaces

```
network interface show
```

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## 14. Data Migration Options

### NetApp to NetApp

#### SnapMirror

Recommended migration method.

Benefits:

- Incremental
  - Fast
  - Efficient
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### Non-NetApp Storage

#### Rsync

Linux migrations.

#### Robocopy

Windows migrations.

## **NetApp XCP**

High-speed migration tool.

## **Cloud Sync**

Cloud-based migration.

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## **Offline Migration**

### **AWS Snow Family**

Suitable for:

- Petabyte-scale migrations
  - Limited bandwidth environments
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# **15. Interview Questions**

## **Q1. What is Amazon FSx for ONTAP?**

A fully managed AWS file storage service built on NetApp ONTAP.

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## **Q2. Is FSx for ONTAP Active-Active?**

No.

Multi-AZ deployments are Active-Passive.

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## **Q3. How is data protected between AZs?**

Synchronous replication.

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## **Q4. What protocols are supported?**

- NFS
  - SMB
  - iSCSI
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## **Q5. What is FlexCache?**

A caching technology that provides low-latency access to remote data.

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**Q6. What is Capacity Pool Tiering?**

Automatic movement of cold data from SSD storage to lower-cost elastic storage.

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**Q7. Can the same data be accessed by Linux and Windows?**

Yes, using ONTAP Multi-Protocol functionality.

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**Q8. What is the preferred migration method from NetApp to FSx?**

SnapMirror.

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**Q9. Which command displays SVMs?**

```
vserver show
```

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**Q10. How long does file system creation typically take?**

Approximately 20–30 minutes.

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## Summary

Amazon FSx for NetApp ONTAP combines the enterprise storage capabilities of ONTAP with the operational simplicity of AWS managed services. It provides:

- Multi-AZ high availability
- Storage efficiency
- Capacity Pool Tiering
- SnapMirror replication
- Multi-protocol access
- Enterprise-grade data protection
- Seamless AWS integration

FSx for ONTAP is an ideal platform for file services, disaster recovery, backup, cloud bursting, application migration, and hybrid cloud deployments.